Focus & Stress Monitor: user manual

How to set up the EEG headphones, what each screen means, when the app asks you to take a break, and what the numbers can and cannot tell you.

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5. When it asks you to take a break
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1. What the app does

The app reads your brain waves through the electrodes in the headphones while you work. It shows three live scores, **Attention**, **Cognitive load** and **Stress**, a chart of your brain-wave bands, and it suggests breaks. Everything is saved so you can look at it later.

All scores are **personal**: they compare you with yourself earlier in the same session, not with other people. A stress score of 80 means "much higher than your own normal level today", not "80% stressed".

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2. What you need

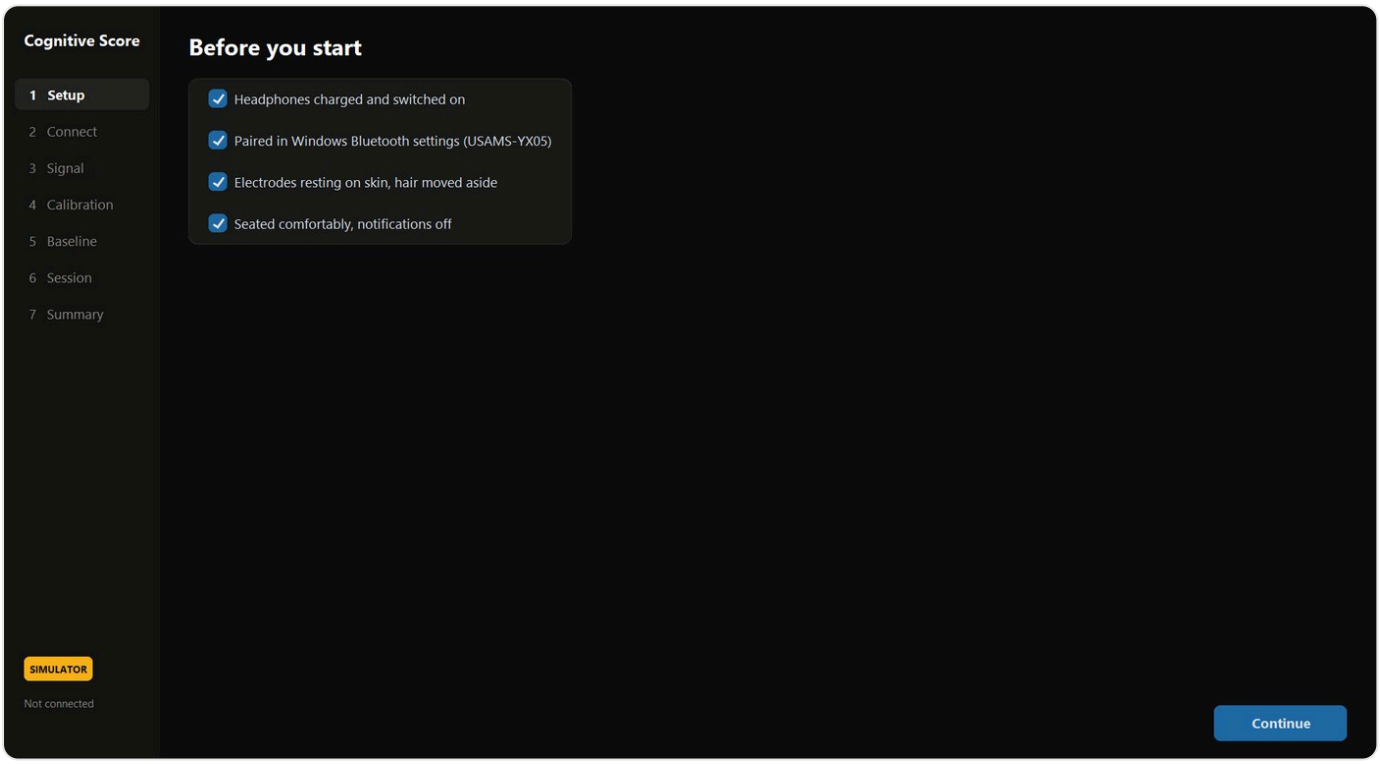
- The Neiry headphones, charged, and paired with Windows (they appear as `USAMS-YX05` in Bluetooth settings).
- This app: double-click `run.bat` in the `cognitive_score` folder.
- About 3 minutes for setup, then as long as you want to work.

3. Step by step

The app walks you through seven steps in order. The list on the left shows where you are. You cannot skip ahead until each step is done, because every later number depends on it.

1 Setup

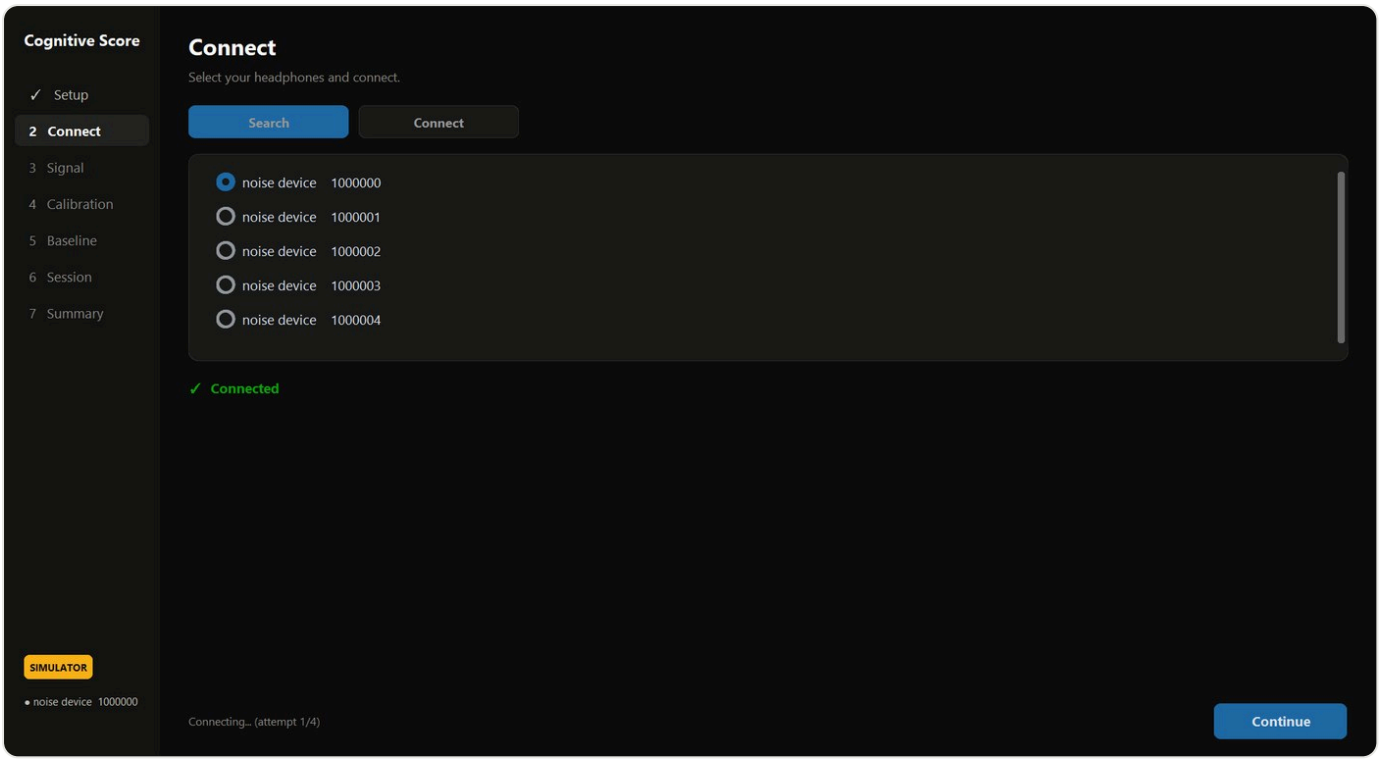
Tick the four checks: headphones charged and on, paired in Windows, electrodes touching skin (move hair away), seated comfortably with notifications off. Then press **Continue**.



Setup: four quick checks before connecting.

2 Connect

Press **Search**. Your headset appears within about 10 seconds. Select it and press **Connect**. If it fails, the app retries three times by itself. If it still fails, switch the headphones off and on and search again.



Connect: search, choose the headset, connect.

3 Signal

Each electrode shows its contact quality. **Good** is below 500 kΩ, **Fair** is up to 1 MΩ, **Poor** is above that. All electrodes must stay Good or Fair for 5 seconds before you can continue. The wavy line is your live brain signal. It should look calm when you sit still.

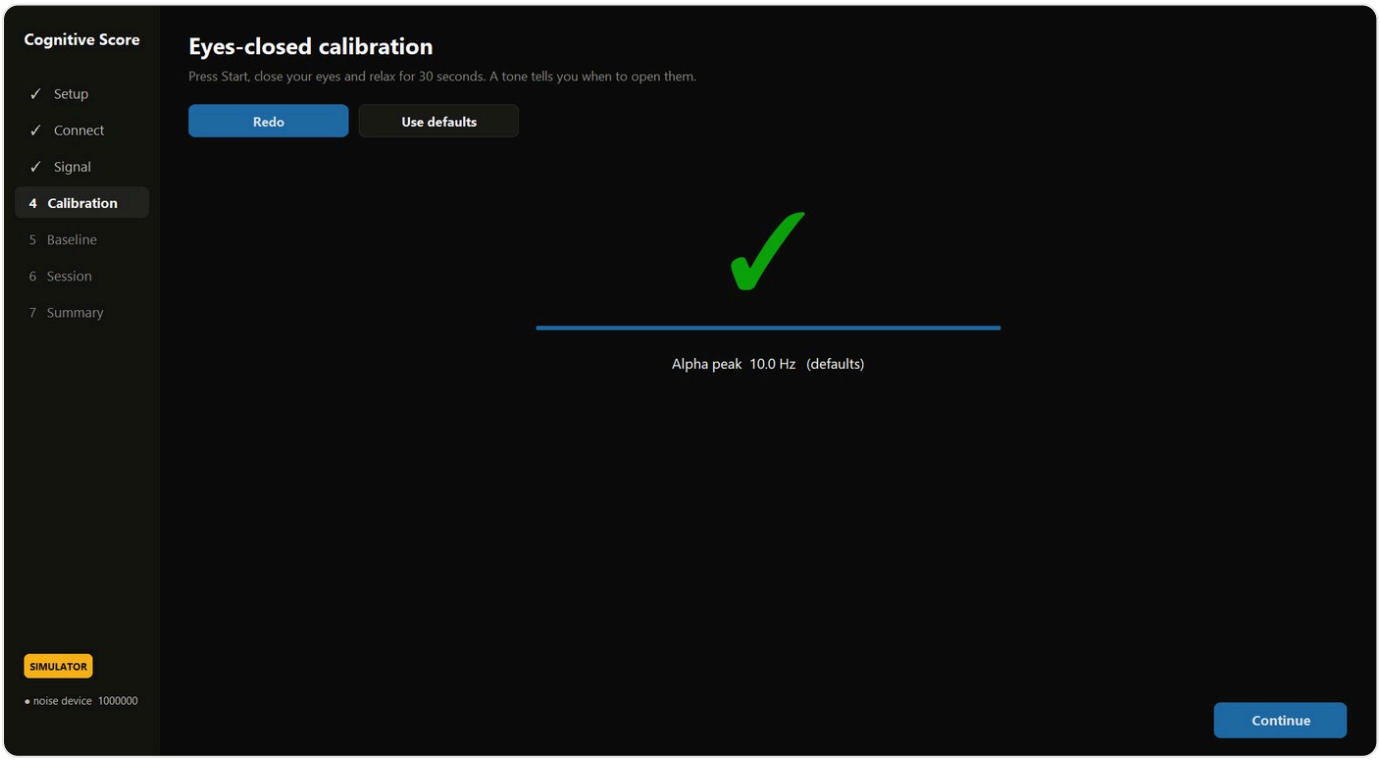
If an electrode stays Poor: press the headphones gently onto your head, move hair away, wait 30 seconds. Dry electrodes get better over the first few minutes as the skin warms up (see the real session). **Skip check** lets you continue anyway, but the scores will be less reliable.



Signal: every electrode needs Good or Fair contact.

4 Calibration (30 seconds, eyes closed)

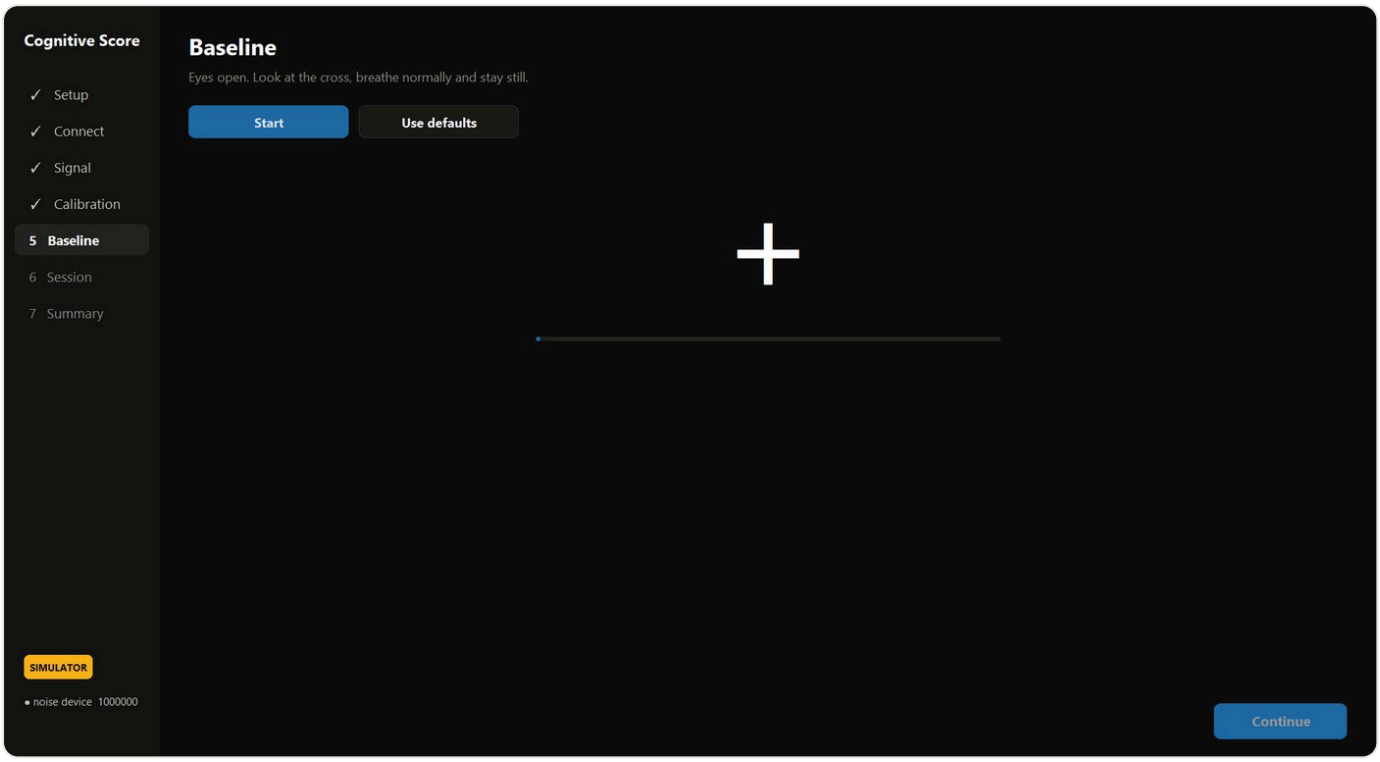
Press **Start**, close your eyes and relax. A tone plays at the start and at the end. The app measures your personal **alpha peak**, the frequency your brain idles at with eyes closed (typically 8–13 Hz). All later calculations are tuned to it.



Calibration: eyes closed for 30 seconds, then your alpha peak is shown.

5 Baseline (30–90 seconds, eyes open)

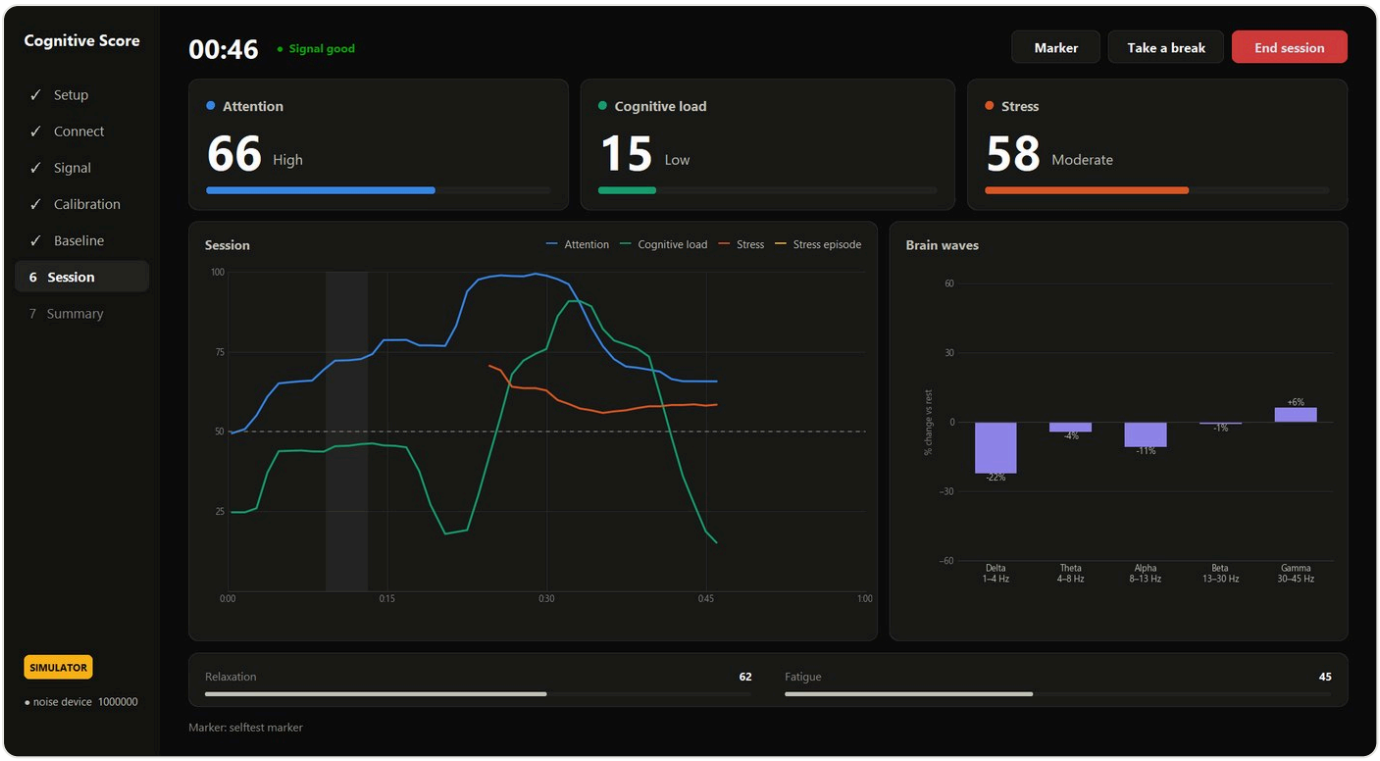
Look at the cross in the middle of the screen and stay still. This is your **rest** level. Attention, cognitive load and the brain-wave chart are compared with it. The app moves on by itself when it is done.



Baseline: eyes open on the cross, keep still.

6 Session

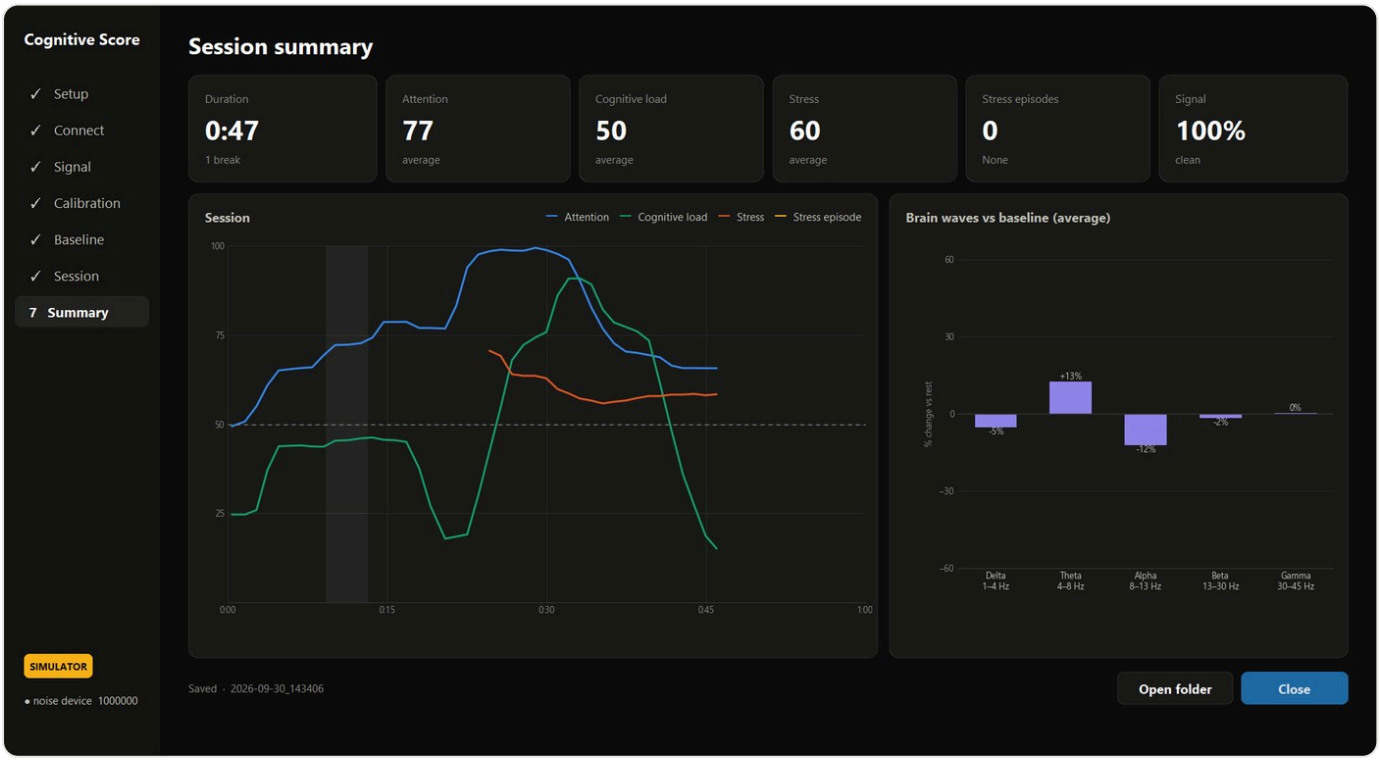
Now just work. The next section explains everything on this screen.



Session screen (from the simulator, so the numbers here are not real brain data).

7 Summary

When you press **End session**, you see your averages, how many stress episodes you had, how clean the signal was, the whole-session timeline and the average brain-wave change. **Open folder** shows the saved files.



Summary after ending the session.

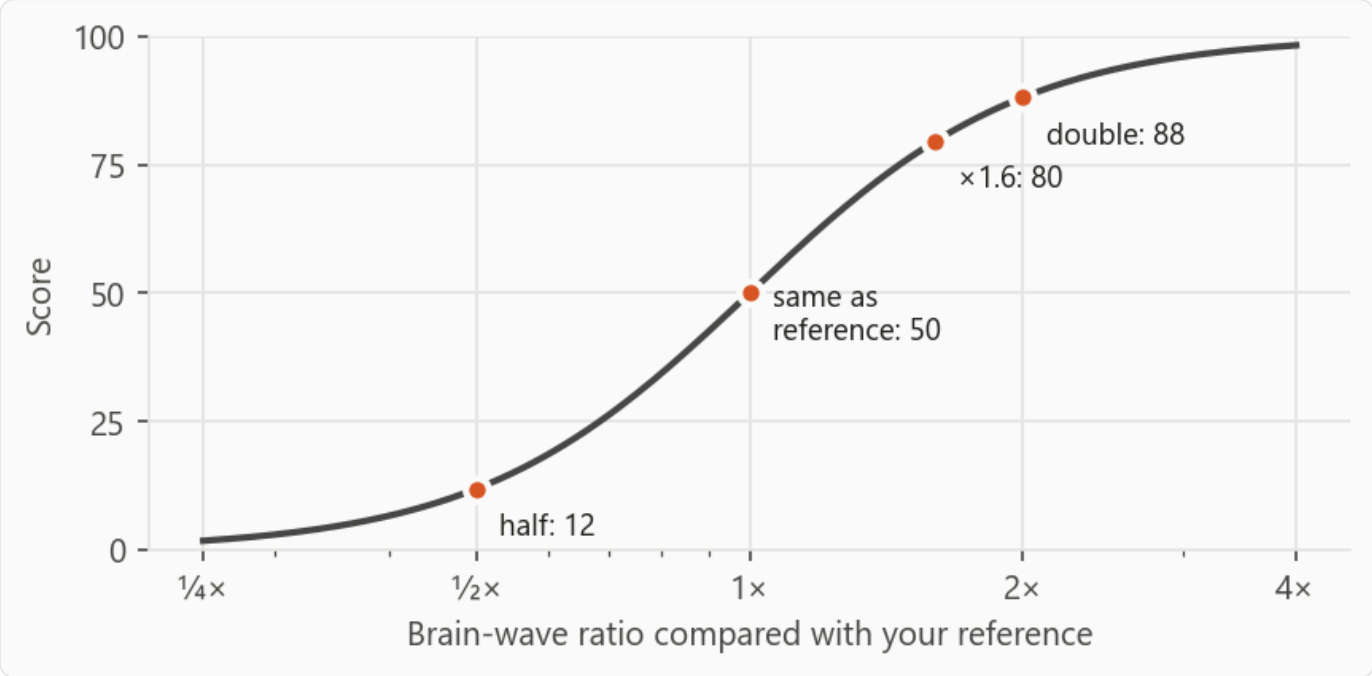
4. Reading the session screen

The three big numbers

Score	What it tracks	Compared with
Attention	Fast (beta) waves relative to idle (alpha) waves. Goes up when you focus.	Your rest baseline (step 5)
Cognitive load	Theta waves relative to alpha. Goes up when a task is mentally demanding (holding many things in mind).	Your rest baseline (step 5)
Stress	High-beta waves (20–30 Hz) relative to alpha: a measure of how "revved up" you are.	Your **first 2 minutes of work**

Each score runs from 0 to 100. Next to the number is a word: **Low** (below 35), **Moderate** (35–64) or **High** (65 and above).

Why stress waits 2 minutes. For the first 2 minutes of the session the Stress card shows "Starts in 1:59" and counts down. During that time the app learns your normal working level. From then on, **50 means "same as your first 2 minutes"**. We changed this after a real session showed that comparing with the eyes-open rest made stress read "High" all the time, just because working at a screen is different from staring at a cross (details in section 7).



How a ratio becomes a score. 50 = same as your reference, double = 88, half = 12. The scale flattens at the ends so one odd reading cannot jump to 0 or 100.

Stress episodes

A **stress episode** starts when stress stays at **80 or above for 30 seconds** (about 1.6 times your working level). The Stress card then shows an orange "Stress episode" tag, you hear one tone, and the timeline is shaded orange. The episode ends when stress stays **below 65 for 30 seconds**. Short spikes are ignored on purpose. Episodes pause during a break.

Session timeline

The three scores over the last 30 minutes, on the same 0–100 scale. The dashed line at 50 is your reference. Grey bands are breaks, orange bands are stress episodes.

Brain waves

Five bars, one per frequency band. Each bar shows how much that band's power has **changed compared with your rest baseline**: +50% means half as much again, −50% means half as much.

Band	Frequency	Usually goes up when…
Delta	1–4 Hz	deep sleep; with dry electrodes, mostly *movement and sweat drift*. Ignore it while awake.
Theta	4–8 Hz	drowsiness, or holding a lot in memory
Alpha	8–13 Hz	relaxing, eyes closed, mind wandering. Drops when you look and focus.
Beta	13–30 Hz	active thinking and focus
Gamma	30–45 Hz	mainly *jaw clenching, chewing, talking and neck tension* at these electrode spots

The bars update every second from the last 5 seconds of clean signal. Movement bursts are removed automatically.

The small meters and the top bar

- **Relaxation**: the vendor's relaxation score (alpha relative to beta), 0–100.
- **Fatigue**: slow waves (theta + alpha) relative to beta, 50 = your first 2 minutes of work. It is smoothed over 1 minute because tiredness builds up slowly. This is the number the break rule watches.
- **Heart rate** appears only on headsets with a pulse sensor.
- **Signal pill**: green "Signal good", or amber "Adjust headphones" when more than 30% of the last 30 seconds had movement or poor contact. While amber, the numbers are greyed and not counted in your averages.

- **Marker**: type a label (for example "started report") to find that moment later in the files.
- **Take a break / Resume** and **End session**.

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5. When it asks you to take a break

A banner appears at the top with a tone and a **Take a break** button. It is triggered by **either** of two rules:

Rule	Exactly when	Why
A. 50-minute timer	50 minutes of work since the session started or since your last real break. The banner says "50 min without a break".	A safe default, even if the EEG cannot see fatigue.
B. Fatigue	The fatigue meter stays at **70 or above for 3 minutes in a row**, and at least 10 minutes have passed since the session started. The banner says "Fatigue has been high for several minutes".	Slow waves grow as you tire. 70 means about 1.4× your level at the start.

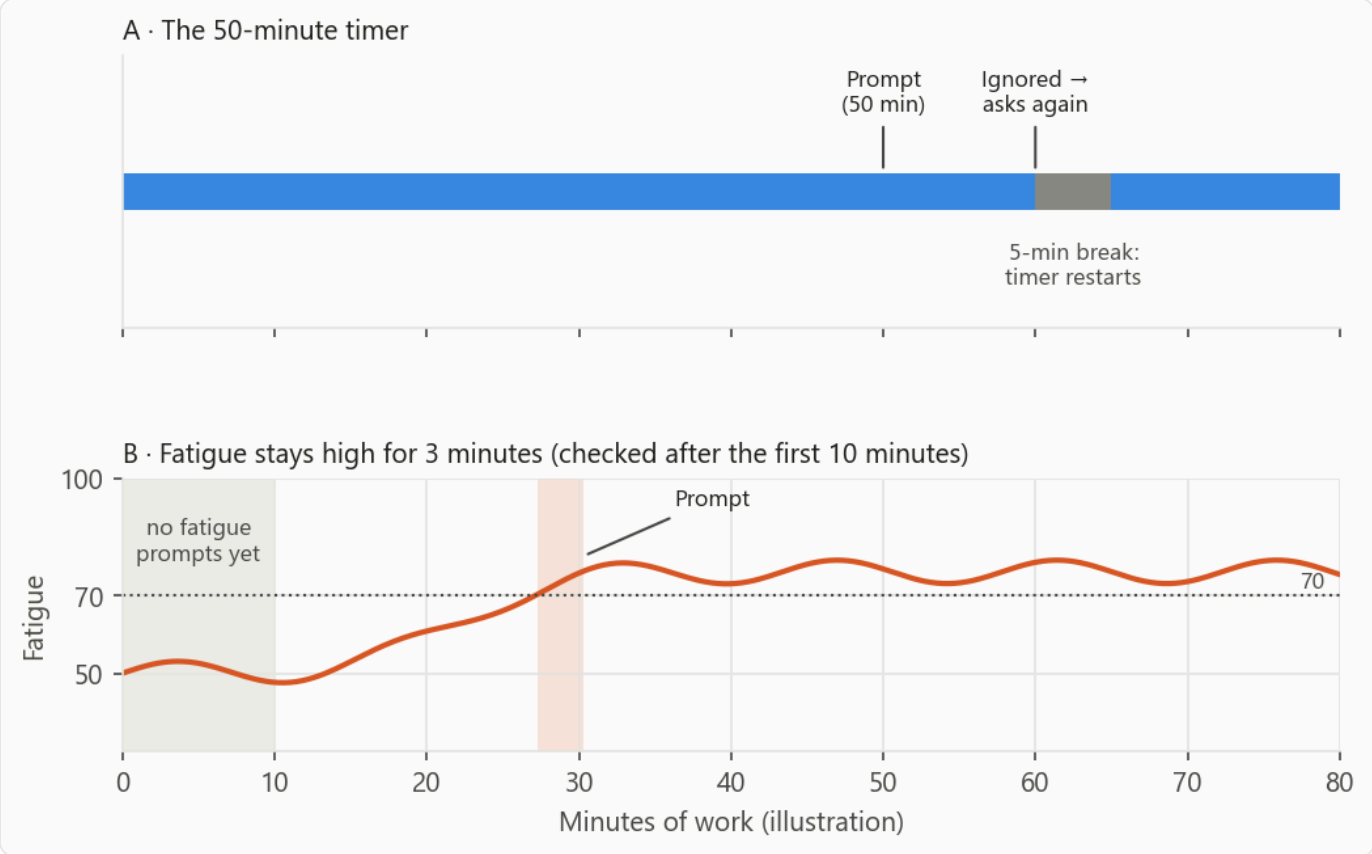


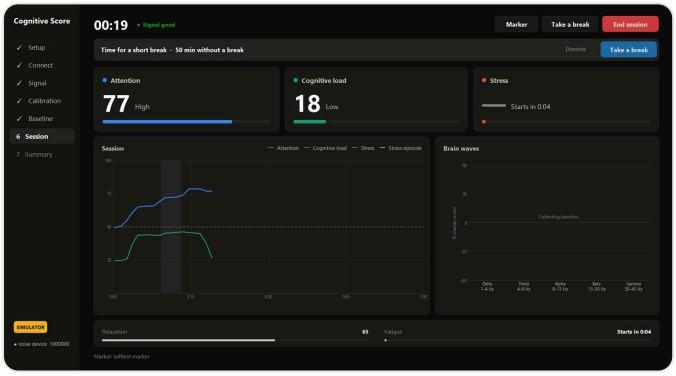
Illustration of the two rules (made-up data). A: the timer, a repeat after an ignored prompt, and a restart after a real break. B: fatigue above 70 for 3 minutes.

The fine print

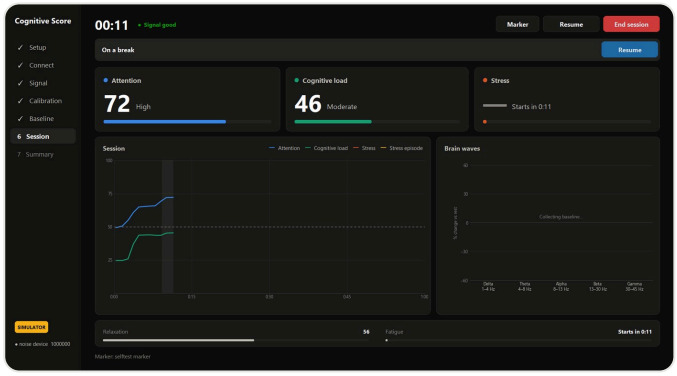
- **At most one prompt every 10 minutes.** If you press Dismiss or ignore it, it comes back 10 minutes later (with the new count, e.g. "60 min without a break").
- **A break must last at least 2 minutes to count.** Shorter breaks are saved, but the 50-minute timer keeps running. (In the real session below, a 1.5-second break would previously have restarted the timer.)
- **No prompts during a break**, and none for 10 minutes after you press Resume.
- **Fatigue during a break does not count.** The 3-minute fatigue clock starts again after you resume.
- No fatigue prompts in the first 10 minutes. The first minutes of a session are not a fair test of tiredness.

How to take a break

Press **Take a break** (top bar or banner). The banner changes to "On a break". Stand up, look away from the screen, move around. You can keep the headphones on. When you come back, press **Resume**. Break time is not counted in your averages and is shown grey on the timeline.



A break prompt.

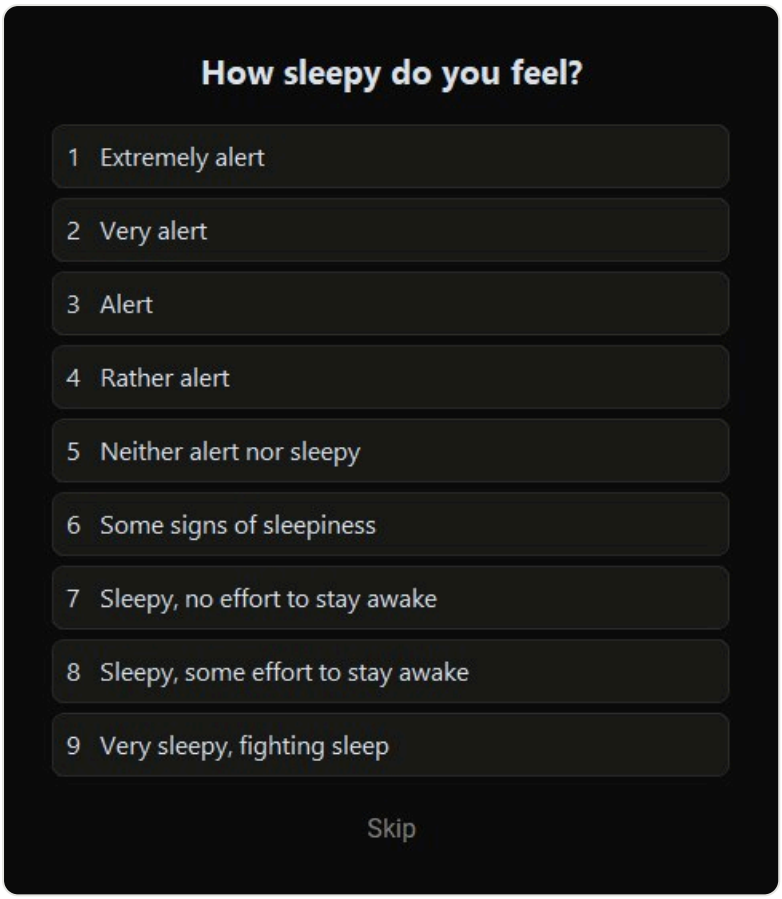


During a break.

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6. Sleepiness check-in

Every 10 minutes a small window asks "How sleepy do you feel?" with a 1–9 scale (the Karolinska Sleepiness Scale, a standard questionnaire). Click the number that fits, or Skip. It does not stop the recording. It does not appear during a break (it waits until you resume). Your answers are saved next to the brain data, so later you can check whether the fatigue meter agrees with how you actually felt.



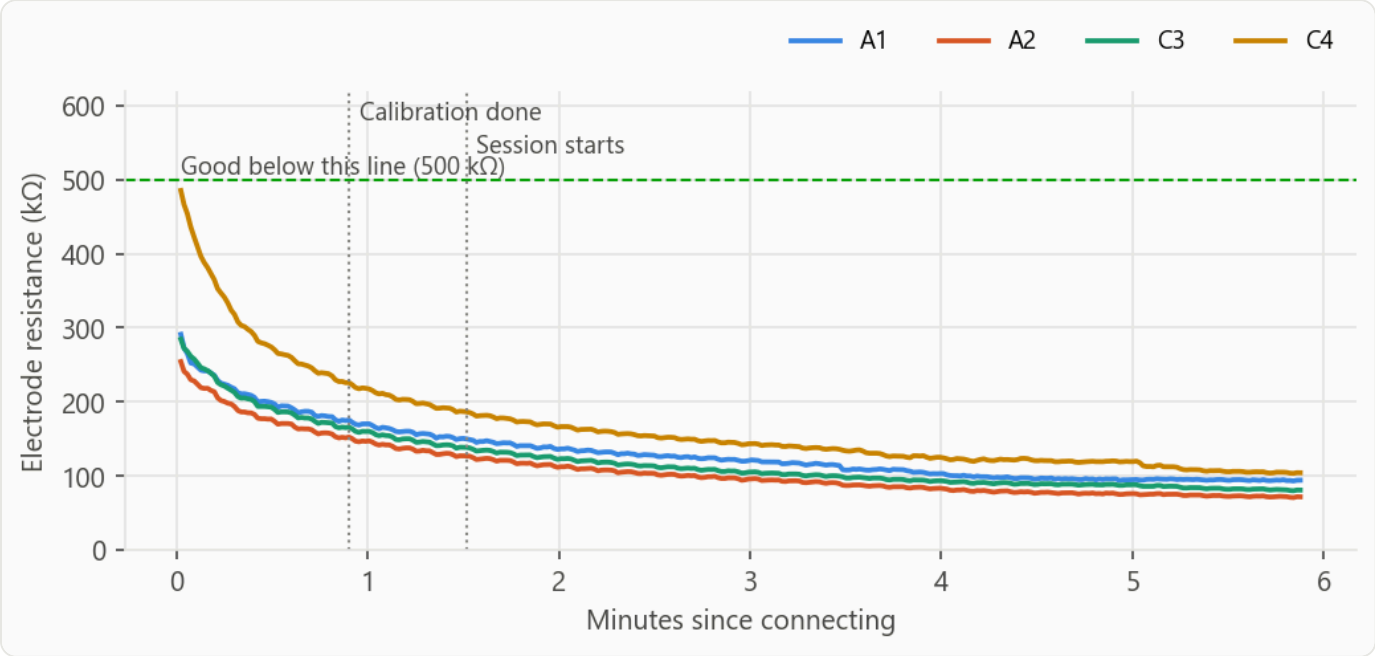
The check-in window.

7. A real session, explained (2026-09-30)

This section uses a real 4.4-minute recording from the headphones (folder `2026-09-30_135243`). The figures show what the **current** version of the app computes from that data. This session is what led to the fixes described below.

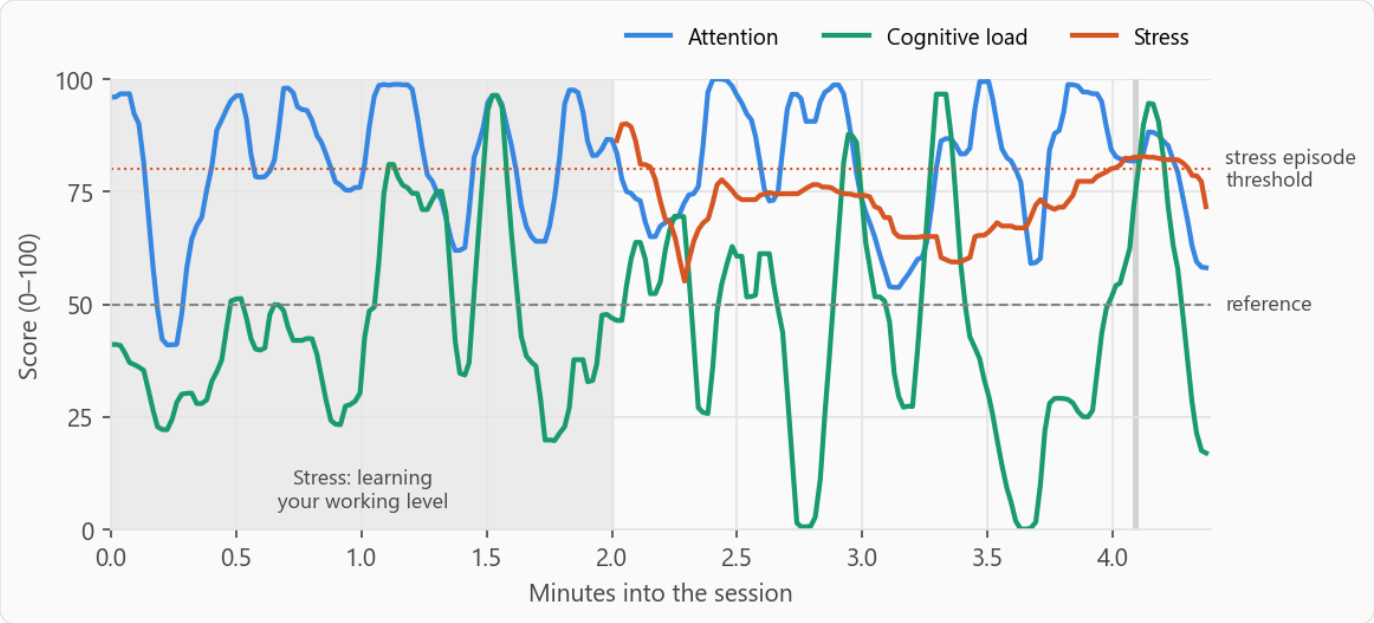
Setup went well

Calibration found an alpha peak of **11.7 Hz** (normal range 8–13 Hz). Only 0.5% of the signal was disturbed by movement. Electrode resistance was already Good at the start (median 177–278 kΩ) and kept falling to 86–129 kΩ during the session as the dry electrodes settled on the skin.



Electrode resistance over the whole session. Lower is better. All four electrodes stayed well inside Good and improved for several minutes.

The scores



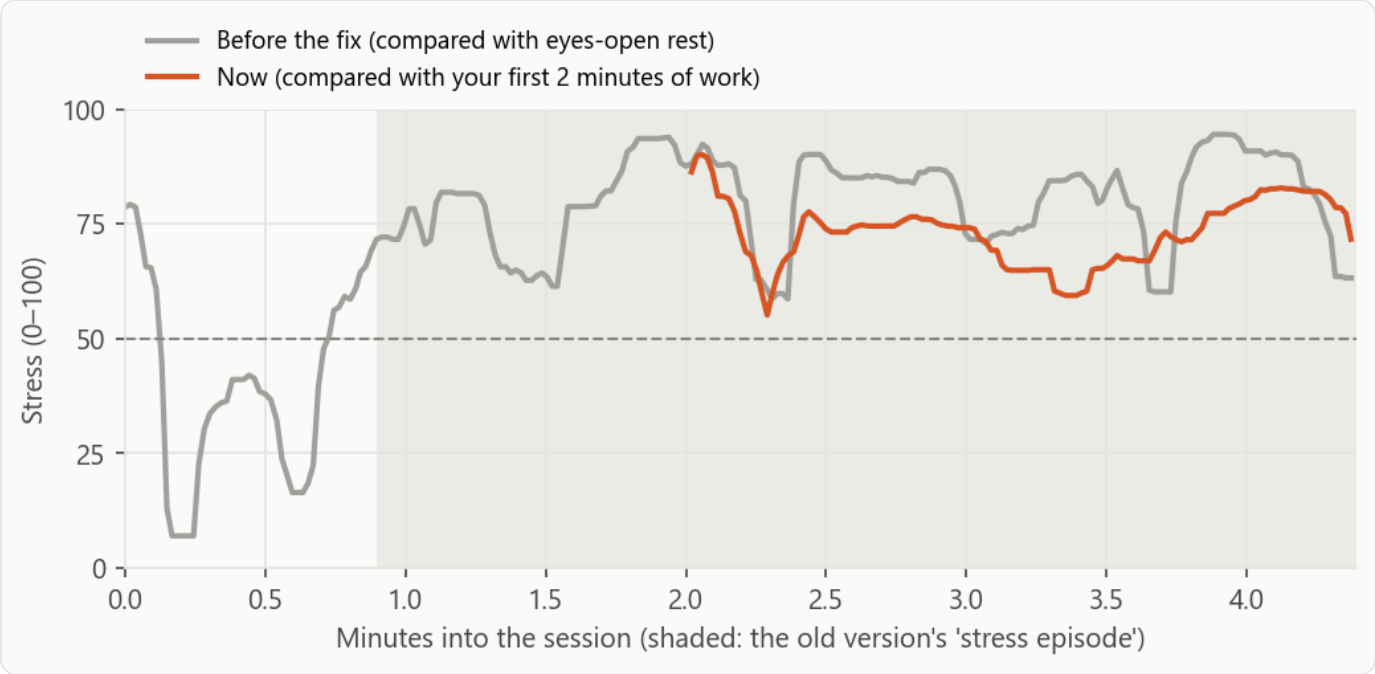
Attention, cognitive load and stress during the session. Stress starts after 2 minutes. The grey band on the left is the 2-minute reference period.

Attention was high throughout (average 81). Cognitive load moved a lot from moment to moment (average 46). Stress, once it started, averaged **73**, with **no stress episodes**.

Fix 1: stress was stuck on "High"

The first version compared stress with the eyes-open rest (step 5). In this session that version showed an average stress of **72** and a single stress episode lasting **3.5 of 4.4 minutes**. The reason: high-beta relative to alpha was about 1.8 times higher while working than while looking at the cross, simply because looking at a busy screen lowers alpha. That is a change of activity, not stress.

Now stress is compared with your **first 2 minutes of work**, smoothed over 30 seconds, and an episode needs 80 or more for 30 seconds. On the same data, that gives **no stress episodes**.

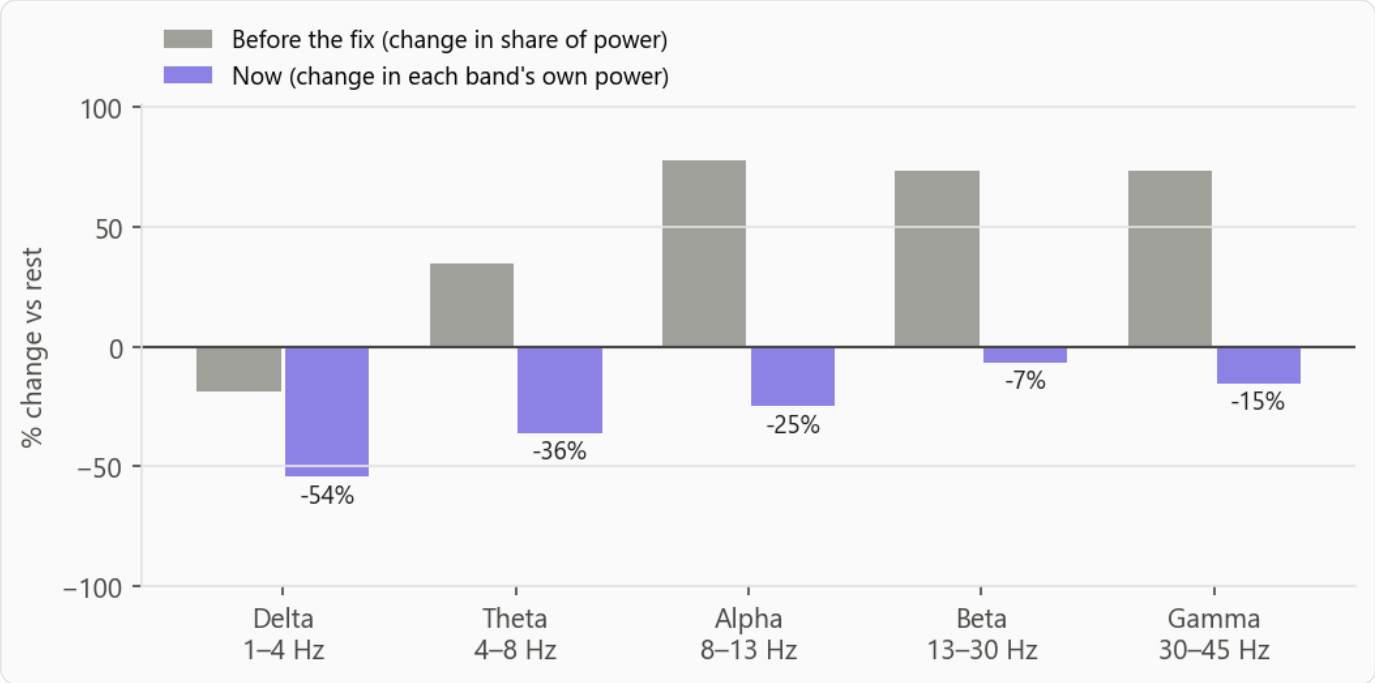


Stress on the same recording, before and after the fix.

Stress still read **High** after minute 2 (about 73), because high-beta relative to alpha kept rising during the session. At the same time the fastest band (gamma) rose by +47% and beta by +33%, while alpha hardly changed (+7%). A broad rise at high frequencies with no alpha change is the typical pattern of **muscle tension** (jaw, neck, frowning) rather than a change in brain state. So in this short test, part of the "stress" was probably muscle. That is still useful: tensing up is part of stress. But check your jaw and shoulders before trusting a high reading (see section 10).

Fix 2: the brain-wave bars all went up together

The first version showed each band's *share* of total power. During the rest baseline the slow drift of the freshly-placed electrodes (delta) made up about three quarters of all power. As the electrodes settled, delta shrank, so every other band's share grew automatically. The old chart showed alpha, beta and gamma all at about +75%. Now each bar compares a band's *own* power with rest, so one band's change cannot move the others.



Average change of each band vs rest for this session, before and after the fix. Now: delta dropped as the electrodes settled, theta and alpha dropped a little (eyes open and busy), beta and gamma stayed about the same.

Fix 3: breaks

In this session you pressed **Take a break** and **Resume** within 1.5 seconds. The old version treated that as a full break and restarted the 50-minute timer. Now a break must last 2 minutes to count. No break prompt would have appeared in this session: it was shorter than 10 minutes (fatigue prompts are off), fatigue stayed between 25 and 54 (below 70), and the 50-minute timer would have first fired at minute **50**.

Also fixed: the check-in no longer pops up during a break, stress episodes end when a break starts, and fatigue measured while on a break no longer counts towards a prompt.

8. Getting a good signal

Problem	What to do
Headset not found	Check it is on and charged; check it is paired in Windows Bluetooth; switch it off and on; search again.
Electrode stays Poor	Move hair aside, press the ear cups gently, wait 30–60 s. A drop of water on the skin helps dry electrodes.
"Adjust headphones" during the session	You moved, talked or clenched your jaw. Sit still for a moment; it clears by itself.
Gamma bar very high	Relax your jaw and shoulders. Gamma at these spots is mostly muscle.
Delta bar very negative or positive	Normal in the first minutes (electrodes settling) or after moving. Ignore delta while awake.
Calibration failed	Try again with eyes fully closed and without moving. The app falls back to typical values if needed.
Headphones disconnected	A red banner appears. Everything up to that point is saved. End the session and start again.

Calibrate every session. Your brain waves and the electrode contact change from day to day; earlier recordings varied up to 7 times between days.

9. What gets saved

Each session gets a folder in `sessions`, named by date and time. **Open folder** on the summary opens it.

File	What is in it
`session.json`	Device, calibration, baselines and the summary you saw at the end
`scores.csv`	The five scores once per second, plus whether the signal was poor or you were on a break
`bands.csv`	Power of each brain-wave band twice per second, and why a window was rejected (movement)
`events.csv`	Markers, breaks, break prompts, stress episodes, check-in answers, signal warnings
other `.csv` files	The raw outputs of the headset software, for later analysis

To rebuild this manual with figures from a newer session: `python docs/build_manual.py --session sessions/<folder>`

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10. What the numbers cannot tell you

> **This is not a medical device.** The scores are estimates from two electrode pairs near the ears.

- **Stress** from brain waves alone is weak evidence. Studies disagree on which brain-wave change means stress. Read it as "more revved up than at the start of this session". Heart-rate based stress (on headsets with a pulse sensor) is stronger.
- **Muscles look like brain waves** above about 20 Hz. Jaw clenching, chewing and talking raise gamma, high beta and therefore stress. Keep your jaw relaxed if you want a clean reading.
- **Scores compare you with yourself today.** Do not compare numbers between people or between days.
- **The break prompt is advice.** The 50-minute timer works even when the EEG cannot see fatigue. Use the check-in answers to see whether the fatigue meter matches how you felt.